CAPSTONE PROJECT REPORT

VoltAI

Intelligent Battery Health Prediction System

SOH & RUL Estimation Using Machine Learning and Deep Learning

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Datasets Used
NASA Battery Cycle-Level Dataset | Oxford Battery Degradation Dataset

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VoltAI: Intelligent Battery Health Prediction System for Electric Vehicles Using Machine Learning and Deep Learning

Abstract—This study presents VoltAI, a comprehensive machine learning pipeline for predicting the State of Health (SOH) and Remaining Useful Life (RUL) of lithium-ion batteries in electric vehicle (EV) applications. Leveraging the NASA Battery Cycle-Level Dataset and the Oxford Battery Degradation Dataset, a unified data framework is established to ensure generalizability across diverse battery chemistries and operating conditions. The study systematically evaluates four model families: Random Forest, XGBoost, Gated Recurrent Unit (GRU), and Bidirectional Long Short-Term Memory with Multi-Head Attention (BiLSTM+Attention). Leakage-safe feature engineering, including cycle-over-cycle differential features, rolling statistics, and interaction terms, is applied to capture temporal degradation trends. Random Forest emerged as the best-performing model, achieving an SOH R^2 of 0.9999 (RMSE = 0.000646) and an RUL R^2 of 0.9982 (RMSE = 45.09 cycles), outperforming all deep learning architectures. An interpretable Battery Health Score on a 300–850 scale, analogous to a credit score, is introduced to translate raw predictions into actionable maintenance insights. SHAP (SHapley Additive exPlanations) analysis confirms that capacity-related rolling features dominate prediction importance, consistent with known electrochemical degradation mechanisms. This work contributes a validated, end-to-end battery health monitoring framework suitable for integration into Battery Management Systems (BMS), fleet monitoring dashboards, and real-world EV applications.

Index Terms—Battery health prognosis, battery management systems (BMS), electric vehicle (EV), lithium-ion battery, remaining useful life (RUL), state of health (SOH), Random Forest, SHAP explainability, deep learning.

Nomenclature

Abbreviations		**Formula Symbols**	
SOH	State of Health	SOH	Predicted State of Health
RUL	Remaining Useful Life	Ct	Capacity at cycle t
EV	Electric Vehicle	C0	Initial battery capacity
BMS	Battery Management System	RULpred	Predicted Remaining Useful Life
RF	Random Forest	Nmax	Maximum cycle number
GRU	Gated Recurrent Unit	RMSE	Root Mean Squared Error
BiLSTM	Bidirectional Long Short-Term Memory	MAE	Mean Absolute Error
MHA	Multi-Head Attention	R^2	Coefficient of Determination
SHAP	SHapley Additive exPlanations	σ	Standard deviation (rolling)
EOL	End of Life	μ	Rolling mean
MAT	MATLAB data format		
EDA	Exploratory Data Analysis		

I. INTRODUCTION

Greenhouse gas emissions driven by fossil fuel combustion pose significant environmental risks, motivating the global transition toward cleaner energy systems [1], [2]. Electric vehicles (EVs), powered by lithium-ion (Li-ion) batteries, represent a primary pathway for decarbonizing road transport. However, the widespread deployment of EVs is constrained by challenges in accurately monitoring battery health, which directly affects vehicle safety, range prediction, and maintenance scheduling.

Accurate estimation of the State of Health (SOH) and Remaining Useful Life (RUL) of Li-ion batteries is critical for effective Battery Management System (BMS) operation. SOH quantifies the current maximum charge capacity relative to the battery's original rated capacity, while RUL estimates the number of charge-discharge cycles remaining before the battery reaches its end-of-life (EOL) threshold. Together, these indicators enable predictive maintenance, optimize charging strategies, and support second-life battery repurposing decisions [3].

Developing accurate SOH and RUL prediction models is inherently challenging due to the nonlinear nature of battery degradation, sensitivity to temperature, charge rate, depth of discharge, and the limited availability of diverse, real-world cycling datasets [3]. Existing approaches span physics-based models, equivalent circuit models, and data-driven machine learning methods. Among data-driven approaches, deep learning architectures—particularly LSTM and Transformer-based models—have demonstrated strong predictive performance [4]–[8]. However, most prior work focuses on either SOH or RUL in isolation, lacks rigorous leakage-safe feature engineering, or has not been validated across heterogeneous datasets.

This work presents VoltAI, a comprehensive machine learning pipeline that addresses these gaps. Key contributions include: (i) a unified data preprocessing framework combining the NASA and Oxford battery datasets; (ii) leakage-safe feature engineering capturing temporal degradation trends; (iii) a systematic comparison of Random Forest, XGBoost, GRU, and BiLSTM+Attention architectures for simultaneous SOH and RUL prediction; (iv) an interpretable Battery Health Score (300–850 scale) for practical deployment; and (v) SHAP-based explainability analysis validating physical consistency of predictions.

II. LITERATURE REVIEW

Machine learning (ML) and deep learning (DL) approaches have demonstrated significant potential in battery performance evaluation, health monitoring, and lifetime estimation. This section reviews recent advancements relevant to the VoltAI framework.

Transformer-based models have been applied to RUL prediction with considerable success. Vaswani et al. [4] introduced the foundational attention mechanism, subsequently adapted by Chen et al. [6] for battery RUL estimation, achieving promising results on the NASA dataset. Wang et al. [7] combined self-attention with Bidirectional LSTM for online RUL prediction, while Nakano et al. [20] proposed a Transformer-based SOH estimator trained on real-world EV trip data achieving an MAE of 0.72% and RMSE of 1.17%.

Ensemble and tree-based methods have also been extensively studied. Swain et al. [24] found that Random Forest outperforms Support Vector Machine for battery RUL prediction after feature selection using one-way ANOVA. Multiobjective arithmetic optimization combined with deep extreme learning machines [10] demonstrated improved RUL accuracy on the NASA dataset. Jafari et al. [17] integrated multilayer perceptrons with support vector regression for enhanced SOH estimation.

Hybrid and multi-task approaches represent the current frontier. Yang et al. [28] proposed the GBLS Booster model combining Broad Learning Systems with CNN-Transformer for joint SOH and RUL estimation. Xia et al. [25] used BiGRU with electrochemical and thermal feature fusion, achieving RMSE below 0.3% on NASA and Oxford datasets. Alsuwian et al. [29] comprehensively reviewed co-estimation frameworks for simultaneous SOH and RUL prediction in BMS applications.

Rolling statistical features and incremental capacity analysis have been shown to improve SOH estimation accuracy. Zhang et al. [15] applied the elbow rule for feature selection with Gaussian process regression, achieving approximately 4% relative error at 80% SOH. Lyu et al. [18] demonstrated that adaptive multi-LSTM architectures using cross-scale relaxation voltage features yield MAEs as low as 1.07%.

Despite these advances, several gaps remain: (i) most methods evaluate on a single dataset, limiting generalizability; (ii) feature engineering protocols rarely address train-test leakage for multi-battery datasets; (iii) model interpretability through methods such as SHAP is infrequently reported. VoltAI addresses all three gaps.

III. BACKGROUND

A. Battery Management System (BMS)

The BMS in EVs monitors and controls key battery parameters including temperature, voltage, current, and state estimations. It functions as a protector, data processor, energy controller, and thermal manager. The BMS computes SOH, SOC, RUL, and State of Power (SOP) to ensure safe, efficient, and reliable battery operation [35].

B. Key Battery Metrics

1) State of Health (SOH):

SOH represents the current maximum charge capacity of a battery relative to its initial rated capacity, expressed as:

$$SOH = C_t / C_0 \times 100\% \quad (1)$$

where C_t is the discharge capacity at cycle t and C_0 is the initial capacity. SOH = 100% at the beginning of life and declines monotonically with cycling. A battery typically reaches EOL when SOH drops below 70–80%.

2) Remaining Useful Life (RUL):

RUL represents the number of charge-discharge cycles remaining before the battery reaches its EOL threshold, defined as:

$$RUL = N_{max} - N_{current} \quad (2)$$

where N_{max} is the maximum cycle number at EOL and $N_{current}$ is the present cycle. RUL prediction enables predictive maintenance and informs replacement scheduling.

C. Performance Metrics

Model performance is assessed using four metrics. The coefficient of determination R^2 measures the proportion of variance explained by the model:

$$R^2 = 1 - SS_{res} / SS_{tot} \quad (3)$$

The Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) quantify prediction magnitude errors:

$$MAE = (1/n) \Sigma|y_i - \hat{y}_i| \quad (4)$$

$$MSE = (1/n) \Sigma(y_i - \hat{y}_i)^2 \quad (5)$$

$$RMSE = \sqrt{[(1/n) \Sigma(y_i - \hat{y}_i)^2]} \quad (6)$$

where y_i is the true value, $\hat{y}_i$ is the predicted value, and n is the total number of samples. Lower RMSE and MAE, combined with higher R^2, indicate superior model performance.

IV. DATASETS

Two publicly available, real-world battery cycling datasets are combined to ensure generalizability across different battery chemistries and testing conditions. Both datasets are aligned to a common schema comprising battery_id, cycle, capacity, voltage, and temperature.

A. NASA Battery Cycle-Level Dataset

The NASA dataset contains cycling data for lithium-ion 18650 cylindrical cells tested at approximately 24 °C ambient temperature. Batteries are labeled B0025, B0049, B0053, and others. Charging follows a constant-current constant-voltage (CCCV) protocol at 1.5 A until 4.2 V, after which voltage is held constant until current drops to 20 mA. Discharge proceeds at constant 2 A current until cell-specific voltage cutoffs. The dataset records temperature, voltage, current, and capacity per cycle, terminating at approximately 70% of initial capacity (EOL). This dataset provides the primary training and testing data for both SOH and RUL prediction tasks.

B. Oxford Battery Degradation Dataset

The Oxford Battery Degradation Dataset contains cycling data from eight small lithium-ion pouch cells, all tested at a controlled temperature of 40 °C maintained by a thermal chamber [34]. Cells undergo CCCV charging followed by discharge under an urban Artemis drive profile. Data collection occurs at intervals of 100 cycles. The increasing battery temperature across successive cycles serves as a key aging indicator [34]. This dataset is used to validate cross-dataset generalization of the trained models.

TABLE I

DATASET COMPARISON: NASA VS. OXFORD BATTERY DEGRADATION DATASETS

Property	NASA Dataset	Oxford Dataset
Format	CSV files	MAT (MATLAB binary)
Battery Type	Li-ion 18650 cylindrical cells	Pouch cells
Key Identifier	battery_id (B0025, B0049, ...)	Cell IDs (Cell1 – Cell8)
Measurements	Cycle, Capacity, Voltage, Temperature	Cycle, Capacity, Voltage, Temperature
Testing Temp.	~24 °C (ambient)	40 °C (controlled chamber)
Charge Protocol	CCCV (1.5 A → 4.2 V)	CCCV charging
Discharge Protocol	Constant current discharge	Urban Artemis drive profile

Property	NASA Dataset	Oxford Dataset
Primary Use	SOH & RUL model training and testing	Cross-dataset generalization

Both datasets are unified into a single DataFrame. Missing voltage and temperature values are recovered through within-battery linear interpolation, followed by forward-fill and backward-fill to preserve temporal continuity.

V. METHODOLOGY

The VoltAI pipeline, illustrated conceptually in Fig. 1, encompasses five stages: data ingestion and cleaning, target construction, feature engineering, leakage-safe data splitting, and multi-model training with comparative evaluation. The following subsections describe each stage in detail.

Fig. 1. VoltAI system pipeline: from raw battery cycling data through feature engineering, model training, Battery Health Score generation, and SHAP explainability.

A. Data Pipeline

Raw NASA CSV files and Oxford MAT (MATLAB binary) files are loaded with automatic filename detection. Column names are normalized and renamed for consistency (e.g., voltage_measured → voltage, temperature_measured → temperature). Both datasets are concatenated into a single unified DataFrame. Missing capacity values are dropped; missing voltage and temperature values are imputed using within-group interpolation. The combined dataset is sorted by battery_id and cycle to preserve strict temporal ordering.

B. Target Construction

SOH targets are computed per battery as the ratio of measured capacity to initial capacity, clipped to [0, 1.2] to handle occasional measurement noise above nominal:

$$SOH_t = C_t / C_0, \quad \text{clipped to } [0, 1.2] \tag{7}$$

RUL targets are computed as the difference between the maximum observed cycle and the current cycle, clipped to zero to prevent negative values:

$$RUL_t = \max(cycle) - cycle_t, \quad \text{clipped} \geq 0 \tag{8}$$

C. Feature Engineering

A total of 17 features are engineered from the raw measurements. Cycle-over-cycle difference features (capacity_change, voltage_change, temp_change) capture instantaneous degradation rate. Rolling 5-cycle mean and standard deviation features (cap_roll_mean_5, volt_roll_mean_5, cap_roll_std_5, etc.) smooth short-term fluctuations and expose trend information. The cycle_ratio feature (cycle / max_cycle) encodes the battery's relative lifecycle position and is included only in the SOH feature set to prevent label leakage. The capacity_ratio feature (capacity / initial_capacity) is included only in the RUL feature set for the same reason. An interaction term (volt_temp_interaction = voltage × temperature) captures coupled thermal-electrical degradation effects, as supported by prior work [18], [25]. An additional capacity_voltage_interaction term is included in the RUL feature set.

TABLE II
ENGINEERED FEATURE SET WITH DESCRIPTIONS

Feature	Description
`cycle`	Charge-discharge cycle number
`voltage`	Terminal voltage measurement (V)
`temperature`	Cell surface temperature (°C)
`capacity_change`	Cycle-over-cycle capacity difference
`voltage_change`	Cycle-over-cycle voltage difference
`temp_change`	Cycle-over-cycle temperature difference
`cap_roll_mean_5`	Rolling 5-cycle mean of capacity
`volt_roll_mean_5`	Rolling 5-cycle mean of voltage
`temp_roll_mean_5`	Rolling 5-cycle mean of temperature
`cap_roll_std_5`	Rolling 5-cycle std. deviation of capacity
`volt_roll_std_5`	Rolling 5-cycle std. deviation of voltage
`temp_roll_std_5`	Rolling 5-cycle std. deviation of temperature
`cycle_ratio`	Cycle / max cycle — SOH feature set only
`capacity_ratio`	Capacity / initial capacity — RUL feature set only
`volt_temp_interaction`	Voltage × Temperature interaction term

Feature	Description
`capacity_voltage_interaction`	Capacity × Voltage interaction term — RUL only

D. Train / Validation / Test Split

GroupShuffleSplit is applied with battery_id as the grouping key to ensure entire batteries appear in only one partition, strictly preventing any form of data leakage across splits. The dataset is divided into approximately 64% training, 16% validation, and 20% test by applying GroupShuffleSplit twice: first to create the train+validation vs. test partition (80/20), then to split train+validation into final training vs. validation (80/20 of the 80%). All scalers are fit exclusively on training data.

E. ML Models

1) Random Forest:

Random Forest is an ensemble of decision trees trained with bootstrap aggregation (bagging). Each tree is grown on a random sample of the training data with a random subset of features considered at each split, reducing variance through averaging [31]. The SOH model uses 300 trees with max depth 12 and min_samples_split = 4; the RUL model uses max depth 14.

2) XGBoost:

XGBoost employs gradient boosted decision trees with second-order Taylor expansion of the loss, L1/L2 regularization, and column subsampling to control overfitting [32]. Both SOH and RUL models use 400 estimators, learning rate 0.03, max depth 6, subsample ratio 0.9, and column subsample ratio 0.9.

3) GRU:

The Gated Recurrent Unit (GRU) is a recurrent architecture designed to capture temporal dependencies via reset and update gates, overcoming the vanishing gradient problem of standard RNNs [33]. The Round 1 architecture uses two GRU layers (64 → 32 units) with Dropout (0.25) and a dense output layer. The improved Round 2 architecture uses 128 → 64 units with Dropout (0.30/0.20) and two dense layers (64, 32 units). Both compile with Huber loss and Adam optimizer.

4) BiLSTM with Multi-Head Attention:

Bidirectional LSTM processes input sequences in both forward and backward directions, enabling each time step to access context from the full sequence history [32]. The attention mechanism assigns learned importance weights to different time steps, allowing the model to focus on critical degradation phases [4], [7]. The Round 1 architecture uses BiLSTM (64 → 32) with 2-head Multi-Head Attention and a residual connection. The improved Round 2 architecture doubles the units (128 → 64) and uses 4-head attention with a lower learning rate of 0.0005. Both use Huber loss and EarlyStopping on validation MAE.

F. Data Augmentation

To increase the effective size of training sequences and improve deep learning model generalization, two augmentation strategies are applied exclusively to training data: (i) Gaussian noise injection with $\sigma = 0.01$ added to each sequence element, and (ii) uniform random scaling of each sequence in the range [0.98, 1.02]. The augmented dataset triples the number of training sequences. No augmentation is applied to validation or test sequences.

G. Battery Health Score

A novel Battery Health Score is introduced to translate raw ML predictions into a business-interpretable metric on a 300–850 scale, designed to be analogous to a financial credit score. The score is computed at battery level as:

$$Score = 300 + 550 \times (0.65 \times SOH_pred + 0.35 \times RUL_ratio) \quad (9)$$

where RUL_ratio = predicted_RUL / battery_max_RUL, clipped to [0, 1] to normalize across batteries with different maximum cycle lengths. The 0.65/0.35 weighting reflects the higher diagnostic importance of SOH relative to RUL for immediate maintenance decisions. Scores are aggregated to battery level by averaging across all test cycles for that battery.

H. SHAP Explainability

SHAP (SHapley Additive exPlanations) values are computed for the final Random Forest SOH model using TreeExplainer, which computes exact Shapley values for tree-based models without approximation [39]. SHAP values assign each feature a contribution score for each prediction, enabling identification of globally important features and local prediction explanations. This analysis validates that model predictions align with known electrochemical degradation mechanisms.

VI. RESULTS AND DISCUSSION

A. Hyperparameter Configuration

Table III summarizes the final hyperparameter configurations for all trained models. Deep learning hyperparameters were tuned based on validation MAE, with the improved Round 2 architectures selected for final reporting.

TABLE III
MODEL HYPERPARAMETER CONFIGURATIONS

Model	SOH Config	RUL Config
Random Forest	n_est=300, depth=12, min_split=4	n_est=300, depth=14, min_split=4
XGBoost	n_est=400, lr=0.03, depth=6, sub=0.9	Same as SOH config
GRU (Round 1)	64→32 units, Drop=0.25, lr=0.001	Same architecture
BiLSTM+Attn (R1)	64→32, MHA 2-head, lr=0.001	Same architecture
Improved GRU	128→64 units, Drop=0.30/0.20, lr=0.0005	Same architecture
Improved BiLSTM	128→64, MHA 4-head, lr=0.0005	Same architecture

B. Deep Learning Hyperparameter Experiments

Prior to final model training, systematic experiments were conducted on the deep learning architectures to identify optimal batch size, learning rate, and batch normalization configuration. Table IV shows that batch size 32 provided the best balance between validation accuracy (83.8%) and training time (36.8 s) compared to batch sizes 16 and 64.

TABLE IV
BATCH SIZE EXPERIMENT RESULTS

Batch Size	Val Acc.	Precision	Recall	Time (s)
16	82.7%	82.1%	81.8%	52.4
32 (Best)	**83.8%**	**83.4%**	**83.0%**	**36.8**
64	82.3%	81.9%	81.5%	28.1

Learning rate experiments (Table V) showed that rate 0.2 yielded the best overall convergence and validation accuracy. Rate 0.3 led to oscillating training loss curves, while rate 0.1 converged more slowly.

TABLE V
LEARNING RATE EXPERIMENT RESULTS

Learning Rate	Train Acc.	Val Acc.	Precision	Time (s)
0.1	83.5%	82.4%	81.9%	38.2
0.2 (Best)	**85.2%**	**83.8%**	**83.1%**	**36.8**
0.3	82.1%	81.0%	80.7%	35.9

Batch normalization experiments (Table VI) demonstrated that applying BN after the dense layer marginally improved validation accuracy to 85.3% and reduced validation loss, compared to no BN or BN before activation. EarlyStopping with patience = 10 provided the best generalization without premature stopping.

TABLE VI
BATCH NORMALIZATION CONFIGURATION COMPARISON

Configuration	Train Acc.	Val Acc.	Val Loss
No Batch Normalization	84.1%	82.9%	0.0312
BN After Dense Layer (Best)	**85.7%**	**85.3%**	**0.0289**
BN Before Activation	84.9%	84.1%	0.0297

C. SOH Prediction Results

Table VII presents SOH prediction results on the held-out test set. Random Forest achieves near-perfect performance with RMSE = 0.000646, MAE = 0.000385, and R^2 = 0.999976, substantially outperforming all other models. XGBoost achieves a strong R^2 of 0.982 but with RMSE more than 27 times larger than Random Forest. Both GRU and the Improved BiLSTM+Attention models underperform relative to the tree-based baselines, with the BiLSTM model yielding a negative R^2 indicative of poor generalization to the test batteries.

TABLE VII
SOH PREDICTION PERFORMANCE — ALL MODELS ON TEST SET

Model	RMSE	MAE	R^2
Random Forest (Best)	**0.000646**	**0.000385**	**0.999976**
XGBoost	0.017574	0.009445	0.982035
GRU	0.126327	0.099945	0.140969
Improved BiLSTM + Attention	0.280867	0.238769	-3.246348

The superior performance of Random Forest is attributable to the highly structured, tabular nature of the engineered features, which are well-suited to tree-based ensemble methods. The deep learning models, which operate on raw sequences and require substantially more training data per battery, are disadvantaged by the relatively small number of batteries across both datasets.

D. RUL Prediction Results

Table VIII presents RUL prediction results. Random Forest again achieves the best performance (RMSE = 45.09 cycles, R^2 = 0.9982), followed by GRU (RMSE = 48.24, R^2 = 0.9931) and XGBoost (RMSE = 120.61, R^2 = 0.9873). The GRU's competitive RUL performance—in contrast to its poor SOH performance—may reflect the sequential nature of the RUL task, where temporal modeling of degradation trajectories provides modest advantage. The BiLSTM model performs poorly across both tasks in its final configuration.

TABLE VIII
RUL PREDICTION PERFORMANCE — ALL MODELS ON TEST SET

Model	RMSE	MAE	R^2
Random Forest (Best)	**45.09**	**17.74**	**0.998223**
GRU	48.24	19.19	0.993126
XGBoost	120.61	32.76	0.987283
Improved BiLSTM + Attention	605.24	212.51	-0.082239

E. Battery Health Score Results

The Battery Health Score system successfully translates quantitative ML predictions into an interpretable, five-band grading scale. Table IX summarizes the score bands. Applied to the test set, batteries in healthy condition (SOH close to 1.0, high RUL ratio) consistently score in the Excellent or Very Good bands (740–850), while batteries near EOL receive Poor scores (300–579). The score formula (Eq. 9) was validated to monotonically decrease with SOH degradation and RUL depletion, confirming physical consistency.

TABLE IX
BATTERY HEALTH SCORE BANDS AND INTERPRETATIONS

Grade	Score Range	Interpretation
Excellent	800 – 850	Battery is in peak condition
Very Good	740 – 799	Minimal degradation observed
Good	670 – 739	Moderate degradation, still healthy
Fair	580 – 669	Noticeable degradation; monitor closely
Poor	300 – 579	Significant degradation; replacement advised

F. SHAP Explainability Analysis

SHAP analysis of the final Random Forest SOH model reveals that capacity-related features dominate prediction importance. Specifically, cap_roll_mean_5 and capacity_change are the two most influential features, reflecting that rolling capacity trends and cycle-to-cycle capacity loss rate are the primary determinants of current battery health. This finding is consistent with electrochemical degradation theory, where capacity fade is driven by lithium plating, SEI layer growth, and active material loss [35].

The cycle_ratio feature ranks third in importance, confirming that the battery's relative lifecycle position is a strong predictor of SOH. The volt_temp_interaction term contributes additively, capturing coupled thermal-electrical degradation effects that are not independently captured by voltage or temperature alone. Raw voltage and temperature features have lower individual importance, suggesting that their temporal derivatives and rolling statistics are more informative than instantaneous measurements. These findings align with prior work on feature importance for battery health prediction [18], [24], [25].

VII. CONCLUSION

A. Conclusion

This work presented VoltAI, a comprehensive data-driven framework for simultaneous SOH and RUL prediction of lithium-ion batteries in EV applications. By combining the NASA and Oxford battery datasets, applying leakage-safe feature engineering, and systematically comparing four model families, VoltAI delivers reliable, physically consistent battery health estimates. Random Forest achieved the best overall performance across both prediction tasks (SOH R^2 = 0.9999, RUL R^2 = 0.9982), outperforming GRU, BiLSTM+Attention, and XGBoost architectures. The novel Battery Health Score (300–850 scale) provides an interpretable, deployment-ready output for BMS integration and fleet monitoring dashboards. SHAP analysis confirmed that capacity-related rolling

features are the most influential predictors, consistent with known electrochemical degradation mechanisms.

B. Limitations

The primary limitation of VoltAI is the use of two benchmark datasets, which—while widely used in the literature—may not fully represent the diversity of real-world battery chemistries (e.g., LFP, NMC), operating temperatures, and drive profiles encountered in deployed EV fleets. The deep learning models, particularly BiLSTM+Attention, underperformed relative to tree-based methods, likely due to the limited number of distinct batteries available for sequential model training. The Battery Health Score weighting coefficients (0.65/0.35) are empirically set and may require recalibration for applications where RUL prediction takes priority over SOH.

C. Future Work

Future work will focus on: (i) extending the dataset to include additional battery chemistries and real-world EV drive profiles to improve generalizability; (ii) investigating physics-informed neural networks (PINNs) that integrate electrochemical degradation models with data-driven learning; (iii) implementing online learning for real-time BMS deployment with incremental model updates; (iv) applying Bayesian uncertainty quantification to provide confidence intervals around SOH and RUL predictions; and (v) model compression techniques (quantization, pruning) to enable efficient onboard inference on resource-constrained embedded BMS hardware.

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